

2009
**HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 84

- Attempt Questions 1–7
- All questions are of equal value

Total marks – 84

Attempt Questions 1–7

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (12 marks) Use a SEPARATE writing booklet.

- (a) Factorise $8x^3 + 27$. **2**
- (b) Let $f(x) = \ln(x - 3)$. What is the domain of $f(x)$? **1**
- (c) Find $\lim_{x \rightarrow 0} \frac{\sin 2x}{x}$. **1**
- (d) Solve the inequality $\frac{x + 3}{2x} > 1$. **3**
- (e) Differentiate $x \cos^2 x$. **2**
- (f) Using the substitution $u = x^3 + 1$, or otherwise, evaluate $\int_0^2 x^2 e^{x^3+1} dx$. **3**

Question 2 (12 marks) Use a SEPARATE writing booklet.

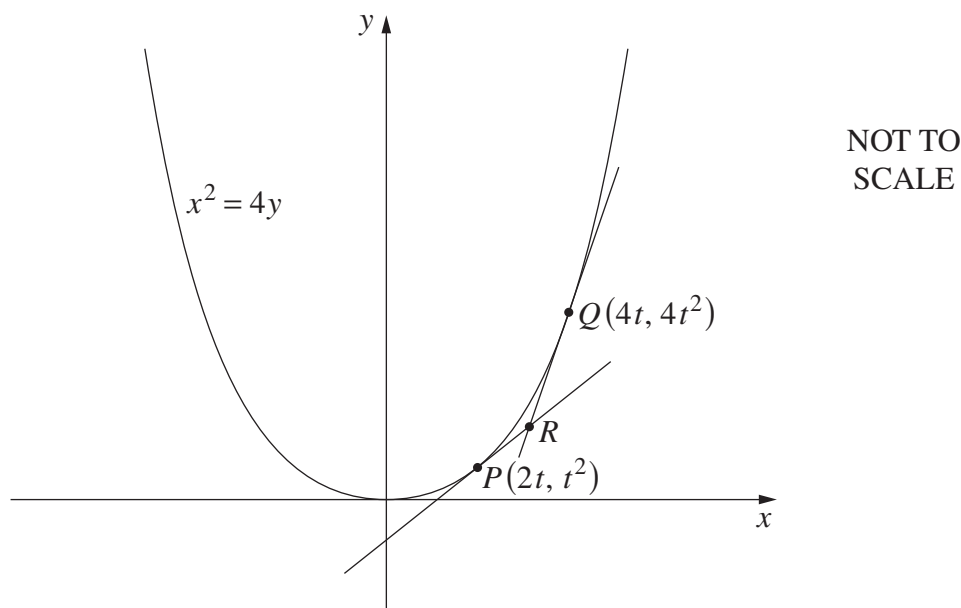
- (a) The polynomial $p(x) = x^3 - ax + b$ has a remainder of 2 when divided by $(x - 1)$ and a remainder of 5 when divided by $(x + 2)$. 3

Find the values of a and b .

- (b) (i) Express $3 \sin x + 4 \cos x$ in the form $A \sin(x + \alpha)$ where $0 \leq \alpha \leq \frac{\pi}{2}$. 2

- (ii) Hence, or otherwise, solve $3 \sin x + 4 \cos x = 5$ for $0 \leq x \leq 2\pi$. Give your answer, or answers, correct to two decimal places. 2

- (c) The diagram shows points $P(2t, t^2)$ and $Q(4t, 4t^2)$ which move along the parabola $x^2 = 4y$. The tangents to the parabola at P and Q meet at R .



- (i) Show that the equation of the tangent at P is $y = tx - t^2$. 2
- (ii) Write down the equation of the tangent at Q , and find the coordinates of the point R in terms of t . 2
- (iii) Find the Cartesian equation of the locus of R . 1

Question 3 (12 marks) Use a SEPARATE writing booklet.

(a) Let $f(x) = \frac{3 + e^{2x}}{4}$.

(i) Find the range of $f(x)$. **1**

(ii) Find the inverse function $f^{-1}(x)$. **2**

(b) (i) On the same set of axes, sketch the graphs of $y = \cos 2x$ and $y = \frac{x+1}{2}$, for $-\pi \leq x \leq \pi$. **2**

(ii) Use your graph to determine how many solutions there are to the equation $2 \cos 2x = x + 1$ for $-\pi \leq x \leq \pi$. **1**

(iii) One solution of the equation $2 \cos 2x = x + 1$ is close to $x = 0.4$. Use one application of Newton's method to find another approximation to this solution. Give your answer correct to three decimal places. **3**

(c) (i) Prove that $\tan^2 \theta = \frac{1 - \cos 2\theta}{1 + \cos 2\theta}$ provided that $\cos 2\theta \neq -1$. **2**

(ii) Hence find the exact value of $\tan \frac{\pi}{8}$. **1**

Question 4 (12 marks) Use a SEPARATE writing booklet.

- (a) A test consists of five multiple-choice questions. Each question has four alternative answers. For each question only one of the alternative answers is correct.

Huong randomly selects an answer to each of the five questions.

- | | | |
|-------|---|----------|
| (i) | What is the probability that Huong selects three correct and two incorrect answers? | 2 |
| (ii) | What is the probability that Huong selects three or more correct answers? | 2 |
| (iii) | What is the probability that Huong selects at least one incorrect answer? | 1 |

- (b) Consider the function $f(x) = \frac{x^4 + 3x^2}{x^4 + 3}$.

- | | | |
|-------|---|----------|
| (i) | Show that $f(x)$ is an even function. | 1 |
| (ii) | What is the equation of the horizontal asymptote to the graph $y = f(x)$? | 1 |
| (iii) | Find the x -coordinates of all stationary points for the graph $y = f(x)$. | 3 |
| (iv) | Sketch the graph $y = f(x)$. You are not required to find any points of inflexion. | 2 |

Question 5 (12 marks) Use a SEPARATE writing booklet.

- (a) The equation of motion for a particle moving in simple harmonic motion is given by

$$\frac{d^2x}{dt^2} = -n^2x$$

where n is a positive constant, x is the displacement of the particle and t is time.

- (i) Show that the square of the velocity of the particle is given by **3**

$$v^2 = n^2(a^2 - x^2)$$

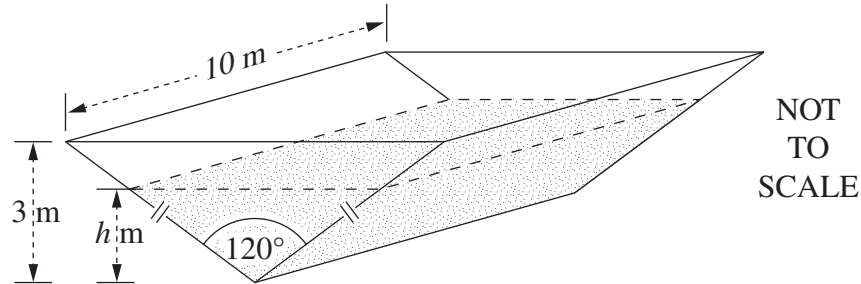
where $v = \frac{dx}{dt}$ and a is the amplitude of the motion.

- (ii) Find the maximum speed of the particle. **1**
- (iii) Find the maximum acceleration of the particle. **1**
- (iv) The particle is initially at the origin. Write down a formula for x as a function of t , and hence find the first time that the particle's speed is half its maximum speed. **2**

Question 5 continues on page 7

Question 5 (continued)

- (b) The cross-section of a 10 metre long tank is an isosceles triangle, as shown in the diagram. The top of the tank is horizontal.



When the tank is full, the depth of water is 3 m. The depth of water at time t days is h metres.

- (i) Find the volume, V , of water in the tank when the depth of water is h metres. 1
- (ii) Show that the area, A , of the top surface of the water is given by 1

$$A = 20\sqrt{3}h.$$

- (iii) The rate of evaporation of the water is given by 2

$$\frac{dV}{dt} = -kA,$$

where k is a positive constant.

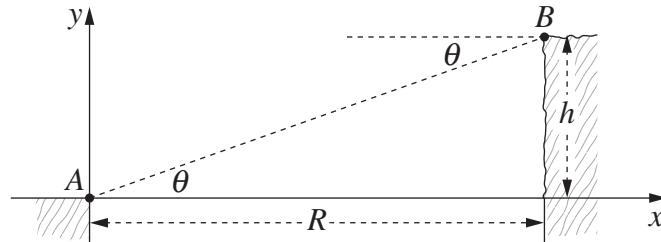
Find the rate at which the depth of water is changing at time t .

- (iv) It takes 100 days for the depth to fall from 3 m to 2 m. Find the time taken for the depth to fall from 2 m to 1 m. 1

End of Question 5

Question 6 (12 marks) Use a SEPARATE writing booklet.

- (a) Two points, A and B , are on cliff tops on either side of a deep valley. Let h and R be the vertical and horizontal distances between A and B as shown in the diagram. The angle of elevation of B from A is θ , so that $\theta = \tan^{-1}\left(\frac{h}{R}\right)$.



At time $t = 0$, projectiles are fired simultaneously from A and B . The projectile from A is aimed at B , and has initial speed U at an angle θ above the horizontal. The projectile from B is aimed at A and has initial speed V at an angle θ below the horizontal.

The equations for the motion of the projectile from A are

$$x_1 = Ut \cos \theta \quad \text{and} \quad y_1 = Ut \sin \theta - \frac{1}{2}gt^2,$$

and the equations for the motion of the projectile from B are

$$x_2 = R - Vt \cos \theta \quad \text{and} \quad y_2 = h - Vt \sin \theta - \frac{1}{2}gt^2.$$

(Do NOT prove these equations.)

- (i) Let T be the time at which $x_1 = x_2$. **1**

Show that $T = \frac{R}{(U + V)\cos\theta}$.

- (ii) Show that the projectiles collide. **2**

- (iii) If the projectiles collide on the line $x = \lambda R$, where $0 < \lambda < 1$, show that **1**

$$V = \left(\frac{1}{\lambda} - 1\right)U.$$

Question 6 continues on page 9

Question 6 (continued)

- (b) (i) Sum the geometric series

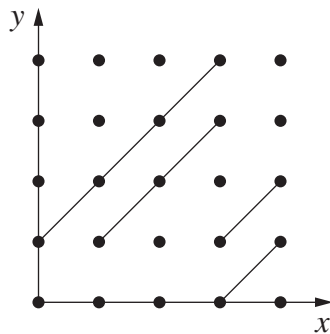
3

$$(1+x)^r + (1+x)^{r+1} + \cdots + (1+x)^n$$

and hence show that

$$\binom{r}{r} + \binom{r+1}{r} + \cdots + \binom{n}{r} = \binom{n+1}{r+1}.$$

- (ii) Consider a square grid with n rows and n columns of equally spaced points.



The diagram illustrates such a grid. Several intervals of gradient 1, whose endpoints are a pair of points in the grid, are shown.

- (1) Explain why the number of such intervals on the line $y = x$ is

1

equal to $\binom{n}{2}$.

- (2) Explain why the total number, S_n , of such intervals in the grid is given by

1

$$S_n = \binom{2}{2} + \binom{3}{2} + \cdots + \binom{n-1}{2} + \binom{n}{2} + \binom{n-1}{2} + \cdots + \binom{3}{2} + \binom{2}{2}.$$

- (iii) Using the result in part (i), show that

3

$$S_n = \frac{n(n-1)(2n-1)}{6}.$$

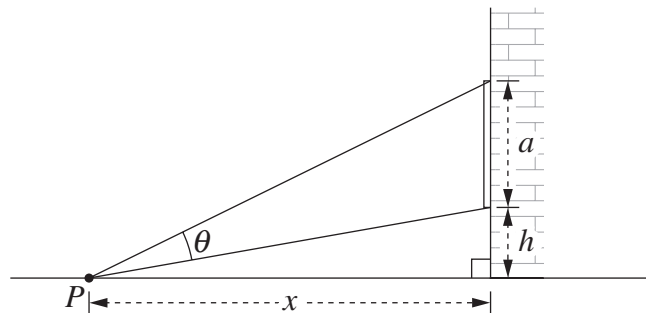
End of Question 6

Question 7 (12 marks) Use a SEPARATE writing booklet.

(a) (i) Use differentiation from first principles to show that $\frac{d}{dx}(x) = 1$. **1**

(ii) Use mathematical induction and the product rule for differentiation to prove that $\frac{d}{dx}(x^n) = nx^{n-1}$ for all positive integers n . **2**

(b) A billboard of height a metres is mounted on the side of a building, with its bottom edge h metres above street level. The billboard subtends an angle θ at the point P , x metres from the building.



(i) Use the identity $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$ to show that **2**

$$\theta = \tan^{-1} \left[\frac{ax}{x^2 + h(a + h)} \right].$$

(ii) The maximum value of θ occurs when $\frac{d\theta}{dx} = 0$ and x is positive. **3**

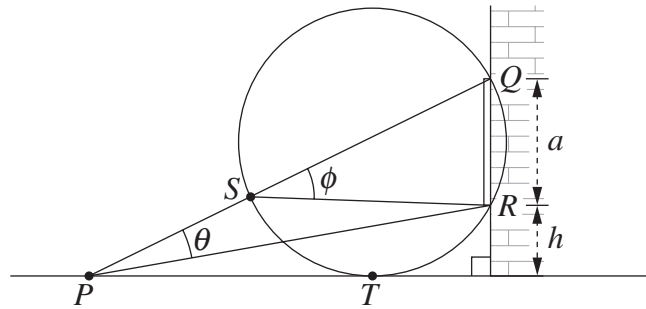
Find the value of x for which θ is a maximum.

Question 7 continues on page 11

Question 7 (continued)

- (c) Consider the billboard in part (b). There is a unique circle that passes through the top and bottom of the billboard (points Q and R respectively) and is tangent to the street at T .

Let ϕ be the angle subtended by the billboard at S , the point where PQ intersects the circle.



Copy the diagram into your writing booklet.

- | | | |
|------|---|----------|
| (i) | Show that $\theta < \phi$ when P and T are different points, and hence show that θ is a maximum when P and T are the same point. | 3 |
| (ii) | Using circle properties, find the distance of T from the building. | 1 |

End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

2009 HSC Mathematics Extension 1 Marking Guidelines

Question 1 (a)

Outcomes assessed: P3

MARKING GUIDELINES

Criteria	Marks
• Correct answer	2
• One of the two factors correct, or equivalent progress	1

Question 1 (b)

Outcomes assessed: P5, H3

MARKING GUIDELINES

Criteria	Marks
• Correct answer	1

Question 1 (c)

Outcomes assessed: H5, H9

MARKING GUIDELINES

Criteria	Marks
• Correct answer	1

Question 1 (d)*Outcomes assessed: PE3***MARKING GUIDELINES**

Criteria	Marks
• Correct solution	3
• Makes significant progress	2
• Recognises an appropriate method	1

Question 1 (e)*Outcomes assessed: PE5, H5***MARKING GUIDELINES**

Criteria	Marks
• Correct solution	2
• Shows an understanding of the product rule or the chain rule	1

Question 1 (f)*Outcomes assessed: HE6***MARKING GUIDELINES**

Criteria	Marks
• Correct answer	3
• Makes the correct substitution including correct limits, or equivalent merit	2
• Attempts the given substitution, or equivalent merit	1

Question 2 (a)

Outcomes assessed: PE3

MARKING GUIDELINES

Criteria	Marks
• Correct solution	3
• Obtains correct equations for a and b , or equivalent progress	2
• Shows some understanding of the remainder theorem	1

Question 2 (b) (i)

Outcomes assessed: HE3

MARKING GUIDELINES

Criteria	Marks
• Correct answer	2
• Correctly obtains A or α , or equivalent progress	1

Question 2 (b) (ii)

Outcomes assessed: HE3

MARKING GUIDELINES

Criteria	Marks
• Correct solution	2
• Obtains a correct result for $x + \alpha$	1

Question 2 (c) (i)

Outcomes assessed: PE4

MARKING GUIDELINES

Criteria	Marks
• Correct solution	2
• Uses the equation of the parabola to show that the gradient at P is t (or equivalent progress)	1

Question 2 (c) (ii)

Outcomes assessed: PE3

MARKING GUIDELINES

Criteria	Marks
• Correct solution	2
• Attempts to solve simultaneously the equations for tangents at P and Q	1

Question 2 (c) (iii)

Outcomes assessed: PE3

MARKING GUIDELINES

Criteria	Marks
• Correct answer	1

Question 3 (a) (i)

Outcomes assessed: P5, H3

MARKING GUIDELINES

Criteria	Marks
Correct answer	1

Question 3 (a) (ii)

Outcomes assessed: HE4

MARKING GUIDELINES

Criteria	Marks
Correct inverse function	2
Makes significant progress	1

Question 3 (b) (i)

Outcomes assessed: H5, H9

MARKING GUIDELINES

Criteria	Marks
Graphs showing important features, including 3 points of intersection	2
Graphs showing some important features	1

Question 3 (b) (ii)

Outcomes assessed: H5, H9

MARKING GUIDELINES

Criteria	Marks
Correct answer (from part (i))	1

Question 3 (b) (iii)*Outcomes assessed: HE3***MARKING GUIDELINES**

Criteria	Marks
• Correct solution	3
• Computes appropriate derivative and shows some understanding of Newton's method	2
• Computes appropriate derivative, or equivalent merit	1

Question 3 (c) (i)*Outcomes assessed: PE6***MARKING GUIDELINES**

Criteria	Marks
• Correct proof	2
• Makes significant progress	1

Question 3 (c) (ii)*Outcomes assessed: H5, HE7***MARKING GUIDELINES**

Criteria	Marks
• Correct answer	1

Question 4 (a) (i)

Outcomes assessed: HE3

MARKING GUIDELINES

Criteria	Marks
• Correct answer	2
• Correct binomial coefficient, or equivalent merit	1

Question 4 (a) (ii)

Outcomes assessed: HE3

MARKING GUIDELINES

Criteria	Marks
• Correct answer	2
• Attempts to add appropriate probabilities, or equivalent merit	1

Question 4 (a) (iii)

Outcomes assessed: HE3

MARKING GUIDELINES

Criteria	Marks
• Correct answer	1

Question 4 (b) (i)

Outcomes assessed: P5, PE2

MARKING GUIDELINES

Criteria	Marks
• Correct solution	1

Question 4 (b) (ii)

Outcomes assessed: H5, HE3

MARKING GUIDELINES

Criteria	Marks
• Correct answer	1

Question 4 (b) (iii)

Outcomes assessed: H6, HE4

MARKING GUIDELINES

Criteria	Marks
• Correct solution	3
• Correct equation i.e. $6x^5 - 12x^3 - 18x = 0$	2
• Attempts to solve $f'(x) = 0$, or equivalent merit	1

Question 4 (b) (iv)

Outcomes assessed: H5, HE7

MARKING GUIDELINES

Criteria	Marks
• Correct graph showing evenness, asymptote and stationary points	2
• Graph showing some important features	1

Question 5 (a) (i)

Outcomes assessed: HE3, HE5

MARKING GUIDELINES

Criteria	Marks
• Correct solution	3
• Makes significant progress	2
• Any statement equivalent to $d\left(\frac{1}{2}v^2\right) = -nx^2$, or equivalent merit	1

Question 5 (a) (ii)

Outcomes assessed: HE3, HE5

MARKING GUIDELINES

Criteria	Marks
• Correct answer	1

Question 5 (a) (iii)

Outcomes assessed: HE3, HE5

MARKING GUIDELINES

Criteria	Marks
• Correct answer	1

Question 5 (a) (iv)

Outcomes assessed: HE3, HE5, HE7

MARKING GUIDELINES

Criteria	Marks
• Correct solution	2
• Correct formula for x , or correct value for t from an incorrect formula	1

Question 5 (b) (i)

Outcomes assessed: P4, PE3, PE6

MARKING GUIDELINES

Criteria	Marks
• Correct solution	1

Question 5 (b) (ii)

Outcomes assessed: HE4, HE7

MARKING GUIDELINES

Criteria	Marks
• Correct solution	1

Question 5 (b) (iii)

Outcomes assessed: HE4, HE7

MARKING GUIDELINES

Criteria	Marks
• Correct solution	2
• Some understanding of related rates, or equivalent merit	1

Question 5 (b) (iv)

Outcomes assessed: HE4, HE7

MARKING GUIDELINES

Criteria	Marks
• Correct answer	1

Question 6 (a) (i)

Outcomes assessed: HE3, HE7

MARKING GUIDELINES

Criteria	Marks
• Correct solution	1

Question 6 (a) (ii)

Outcomes assessed: HE3, HE7

MARKING GUIDELINES

Criteria	Marks
• Correctly shows that $y_1 = y_2$ at time T	2
• Attempts to show that $y_1 = y_2$ at time T , or equivalent merit	1

Question 6 (a) (iii)

Outcomes assessed: HE3, HE7

MARKING GUIDELINES

Criteria	Marks
• Correct solution	1

Question 6 (b) (i)

Outcomes assessed: H5, HE2, HE3

MARKING GUIDELINES

Criteria	Marks
• Correct solution	3
• Makes significant progress	2
• Correctly sums the geometric series, or equivalent merit	1

Question 6 (b) (ii) (1)

Outcomes assessed: HE3

MARKING GUIDELINES

Criteria	Marks
• Correct explanation	1

Question 6 (b) (ii) (2)

Outcomes assessed: HE3

MARKING GUIDELINES

Criteria	Marks
• Correct explanation	1

Question 6 (b) (iii)

Outcomes assessed: HE3, HE7

MARKING GUIDELINES

Criteria	Marks
• Correct solution	3
• Uses the result in part (i) to make substantial progress	2
• Writes S_n in a form allowing use of the result in part (i), or equivalent merit	1

Question 7 (a) (i)

Outcomes assessed: P7

MARKING GUIDELINES

Criteria	Marks
• Correct solution	1

Question 7 (a) (ii)

Outcomes assessed: HE2

MARKING GUIDELINES

Criteria	Marks
• Correct solution	2
• Establishes the inductive step, or equivalent merit	1

Question 7 (b) (i)

Outcomes assessed: PE6, HE3, HE7

MARKING GUIDELINES

Criteria	Marks
• Correct solution	2
• Attempts to use the identity to find θ	1

Question 7 (b) (ii)

Outcomes assessed: H5, HE4, HE7

MARKING GUIDELINES

Criteria	Marks
• Correct solution	3
• Demonstrates understanding of the chain rule for differentiation and attempts to solve $\frac{d\theta}{dx} = 0$	2
• Demonstrates understanding of the chain rule for differentiation	1

Question 7 (c) (i)

Outcomes assessed: P4, PE2, PE3

MARKING GUIDELINES

Criteria	Marks
• Correct solution	3
• Any two of the observations below, or equivalent merit	2
• Observes that $\phi = \theta + \angle SRP$ OR • Observes that ϕ remains constant as P moves OR • Observes that $\phi = \theta$ when $P = T$ OR • Equivalent merit	1

Question 7 (c) (ii)

Outcomes assessed: PE3

MARKING GUIDELINES

Criteria	Marks
• Correct solution, showing evidence of using the correct circle property	1

Mathematics Extension 1

2009 HSC Examination Mapping Grid

Question	Marks	Content	Syllabus outcomes
1 (a)	2	1.3	P3
1 (b)	1	4.1, 12.3	P5, H3
1 (c)	1	13.4	H5, H9
1 (d)	3	1.4	PE3
1 (e)	2	8.8, 13.5	PE5, H5
1 (f)	3	11.5	HE6
2 (a)	3	16.2	PE3
2 (b) (i)	2	5.9, 13.1	HE3
2 (b) (ii)	2	5.9, 13.3	HE3
2 (c) (i)	2	9.6	PE4
2 (c) (ii)	2	9.6	PE3
2 (c) (iii)	1	9.6	PE3
3 (a) (i)	1	4.1, 12.3	P5, H3
3 (a) (ii)	2	15.1	HE4
3 (b) (i)	2	13.3	H5, H9
3 (b) (ii)	1	13.3	H5, H9
3 (b) (iii)	3	16.4	HE3
3 (c) (i)	2	5.7	PE6
3 (c) (ii)	1	13.1	H5, HE7
4 (a) (i)	2	3.3, 18.2	HE3
4 (a) (ii)	2	18.2	HE3
4 (a) (iii)	1	18.2	HE3
4 (b) (i)	1	4.2	P5, PE2
4 (b) (ii)	1	10.5	H5, HE3
4 (b) (iii)	3	10.2	H6, HE4
4 (b) (iv)	2	10.5	H5, HE7
5 (a) (i)	3	14.4	HE3, HE5

Question	Marks	Content	Syllabus outcomes
5 (a) (ii)	1	14.4	HE3, HE5
5 (a) (iii)	1	14.4	HE3, HE5
5 (a) (iv)	2	14.4	HE3, HE5, HE7
5 (b) (i)	1	5.1, 5.5	P4, PE3, PE6
5 (b) (ii)	1	14.2	HE4, HE7
5 (b) (iii)	2	14.2	HE4, HE7
5 (b) (iv)	1	14.2	HE4, HE7
6 (a) (i)	1	14.3	HE3, HE7
6 (a) (ii)	2	14.3	HE3, HE7
6 (a) (iii)	1	14.3	HE3, HE7
6 (b) (i)	3	7.3, 17.3	H5, HE2, HE3
6 (b) (ii) (1)	1	18.1	HE3
6 (b) (ii) (2)	1	18.1	HE3
6 (b) (iii)	3	18.1	HE3, HE7
7 (a) (i)	1	8.8	P7
7 (a) (ii)	2	7.4, 8.8	HE2
7 (b) (i)	2	5.7, 15.2	PE6, HE3, HE7
7 (b) (ii)	3	10.6, 15.5	H5, HE4, HE7
7 (c) (i)	3	2.3, 2.10	P4, PE2, PE3
7 (c) (ii)	1	2.10	PE3